Forecasting Conformal Coverage Loss under Model Compression

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Abstract

Deployed classifiers are routinely compressed after calibration: a model whose conformal guarantee was calibrated in FP32 ships with quantized weights. We show empirically that this silently invalidates the guarantee, and, more usefully, that the coverage loss is *forecastable before deployment*. Treating post-training quantization as a computable shift of the nonconformity-score distribution, an elementary Kolmogorov/total-variation argument upper-bounds the coverage deficit under stale calibration by the distance between the reference and deployed score distributions, both measurable on the labeled calibration set already in hand. Across 2 datasets $\times$ 2 architectures $\times$ 5 bit-widths (60 runs), the bound held in every run, and in the damage regime on peaked score distributions it is nearly tight: at 4-bit and 3-bit precision the measured deficit matched the total-variation distance to within 0.003. We further characterize what recalibration can and cannot repair: validity is restored at every precision tested, including models degraded to 47.8% accuracy, while informativeness collapses past an architecture-dependent precision knee (mean prediction-set size 0.91 at 8-bit versus 5.6–9.4 classes at 3-bit). A deployment corollary under simulated sensor drift shows that refreshing only the calibration scores (kilobytes; no retraining) maintains validity where launch calibration decays from 90.7% to 20.9% coverage, with prediction-set growth acting as the model-health signal. All results are reproducible from public code and raw data.

1 Introduction

Split conformal prediction wraps any classifier with a distribution-free guarantee: prediction sets that contain the true label with probability at least $1-\alpha$ under exchangeability [1, 2, 3]. The guarantee is calibrated on held-out data using the model's nonconformity scores. In deployment practice, however, the model that ships is rarely the model that was calibrated. Post-training quantization (PTQ), pruning, and distillation intervene between calibration and deployment, particularly in resource-constrained settings (mobile, embedded, onboard spacecraft) where compression is mandatory.

This paper asks a narrow question with practical consequences: *what does compression do to a conformal guarantee calibrated on the uncompressed model, and can the damage be predicted in advance?* Our answers:

1. **Stale calibration silently undercovers** (Section 4.1). Reusing FP32 calibration on an INT8 deployment undercovers at every miscoverage level tested, while accuracy is essentially unchanged, the failure is invisible to accuracy monitoring.

2. **The loss is forecastable** (Sections 3, 4.3). The coverage deficit is upper-bounded by the total-variation (TV) distance between reference and deployed score distributions on the calibration

set — a quantity computable on the ground, before deployment, with no new data. The bound is elementary; the contribution is empirical: it held in 60/60 runs across datasets, architectures, and bit-widths, and in the damage regime on peaked score distributions it is nearly an equality.

3. **Recalibration restores validity but not informativeness** (Section 4.4). Coverage returns to target at every precision, even for badly damaged models; mean prediction-set size, by contrast, exhibits a sharp knee in precision, one full bit earlier for MobileNetV3 than for ResNet18. Set size on calibration data is therefore the natural acceptance test for a compressed artifact.

4. **Deployment corollary** (Section 5): under progressive sensor drift, periodically refreshing the calibration scores (200 labeled samples per refresh; no retraining) largely preserves validity, and set-size growth signals when the model itself, rather than its calibration, requires intervention.

2 Related work

Conformal prediction under shift. Barber et al. [4] bound the coverage gap of conformal prediction beyond exchangeability by the total variation between calibration and test distributions, with weighted variants; subsequent work refines such bounds under Wasserstein and other metrics. Our setting is complementary: the data remain exchangeable, and the shift is induced in the *score function* by compressing the model. We instantiate the same style of bound for this self-inflicted, fully measurable shift and characterize its tightness empirically.

Uncertainty for onboard and remote-sensing models. Marchante Arjona et al. [5] study conformal prediction for onboard Earth observation under domain shift with weighted conformal methods, and the same group proposes class-conditional randomized smoothing against structured spectral corruptions [6]. These address input-side shift; neither considers compression of the model itself, which is the axis studied here. We view the two as composable layers of the same deployment problem.

Compression and uncertainty. A recent benchmark studies uncertainty under compression for large language models [7], reporting threshold-like degradation. Our vision results differ: on EuroSAT and CIFAR-10 the collapse of stale-calibration coverage is smooth and tracks the measured score shift (Section 4.2), and we additionally study the stale-versus-recalibrated distinction and set-size efficiency, which that benchmark does not address.

3 Setup and the forecasting bound

Let f_{ref} be the calibrated (reference) classifier and f_q its compressed deployment. Split conformal prediction with the LAC score $s(x, y) = 1 - p_y(x)$ computes the threshold $\hat{q}$ as the $\lceil (1-\alpha)(n+1) \rceil / n$ empirical quantile of calibration scores $S_i^{\text{ref}} = s_{\text{ref}}(X_i, Y_i)$, giving $\mathbb{P}(S^{\text{ref}} \le \hat{q}) \ge 1-\alpha$ on exchangeable data. If deployment instead evaluates f_q while retaining $\hat{q}$ (*stale calibration*), realized coverage is $\mathbb{P}(S^q \le \hat{q})$, and the deficit obeys

$$(1-\alpha) - \mathbb{P}(S^q \le \hat{q}) \;\le\; \sup_t |F_{S^{\text{ref}}}(t) - F_{S^q}(t)| \;=\; d_K(S^{\text{ref}}, S^q) \;\le\; d_{TV}(S^{\text{ref}}, S^q), \tag{1}$$

up to the usual finite-sample slack in the calibration quantile. The inequality itself is elementary; its value is operational. Because compression is applied by the deployer, both score distributions are computable on the *same labeled calibration set*: quantize, re-score, estimate d_{TV} (we use 1000-bin

histograms on $[0, 1]$). The result is a pre-flight forecast: an upper bound on the coverage that will be lost if calibration is not refreshed. The empirical questions are whether the bound holds under estimation error and how tight it is; the remainder of the paper measures this.

4 Experiments

Protocol. Datasets: EuroSAT RGB [8] (27,000 tiles, 10 classes; 70/15/15 train/calibration/test) and CIFAR-10 [9]. Architectures: ResNet18 [10] and MobileNetV3-Small [11], ImageNet-initialized, 12 epochs AdamW. Compression: symmetric per-channel weight quantization to $b \in \{8, 6, 4, 3, 2\}$ bits (simulated PTQ: quantized values, FP32 arithmetic), plus production INT8 (FX-graph PTQ) for Section 4.1. Multi-seed throughout; error bars are standard deviations over seeds. Code, raw results, and exact environment are public; a fresh-clone reproduction test recovers the reported numbers exactly.[1]

4.1 Stale calibration silently undercovers

On EuroSAT/ResNet18 with production INT8 PTQ (20 seeds), accuracy is essentially unchanged (98.13% → 98.03%) while stale-calibration coverage falls below target at every level tested: 88.8±0.8% against a 90% target (α=0.10), 94.4±0.6% against 95%, and 83.0±0.8% against 85%, the gap growing with α. Recalibrating on the quantized model restores target coverage at all levels ($90.0 \pm 0.7\%$, $94.9 \pm 0.5\%$, $84.9 \pm 0.6\%$). The practical rule is stark: *calibrate the artifact that ships, not the model that trained.*

4.2 Coverage collapse is smooth and tracks the score shift

Sweeping precision on EuroSAT/ResNet18 (5 seeds; Figure 1): stale-calibration coverage falls monotonically — 89.8% (8-bit), 88.9% (6-bit), 66.5% (4-bit), 11.6% (3-bit), 1.9% (2-bit) against a 90% target, tracking the relative weight perturbation (1.1%, 4.4%, 19.6%, 42.2%, 75.2%) and the mean absolute probability shift. We observe no threshold effect: degradation is gradual in the measured shift, in contrast to the threshold-like behaviour reported for LLMs [7]. Recalibration restores target coverage at *every* precision, including a 3-bit model at 47.8% accuracy (coverage 89.7%) and a 2-bit model at 10.3% accuracy (coverage 100%, attained by prediction sets growing toward all classes, see Section 4.4).

4.3 The bound holds and is nearly tight in the damage regime

Figure 2 plots measured deficit against estimated $d_{TV}(S^{\text{ref}}, S^q)$ per bit-width (5 seeds). At 8- and 6-bit the bound is conservative (deficit 0.002 and 0.011 against TV 0.030 and 0.034). In the damage regime it is nearly an equality: deficit 0.235 against TV 0.238 at 4-bit and 0.784 against 0.783 at 3-bit. The single nominal violation (0.001 at 3-bit) is within histogram-estimation error of the TV distance. Thus the forecast errs safe where damage is small and is near-exact where it matters.

Generality. Repeating the test over the full grid, {EuroSAT, CIFAR-10} × {ResNet18, MobileNetV3-Small} × {8,6,4,3,2}-bit, 3 seeds each, the bound of Eq. (1) held in **60 of 60 runs** (Figure 3). Tightness is systematic rather than accidental: on EuroSAT, whose score distribution is strongly peaked (high-confidence task), deficit approaches TV (e.g. 0.776 vs. 0.797, ResNet18 at 3-bit); on CIFAR-10, with broader score distributions, the bound is conservative (0.371 vs. 0.525). The forecast therefore never underestimated the loss in our experiments and approaches equality

[1] https://github.com/ss1738/euroSAT-CNN-landcover

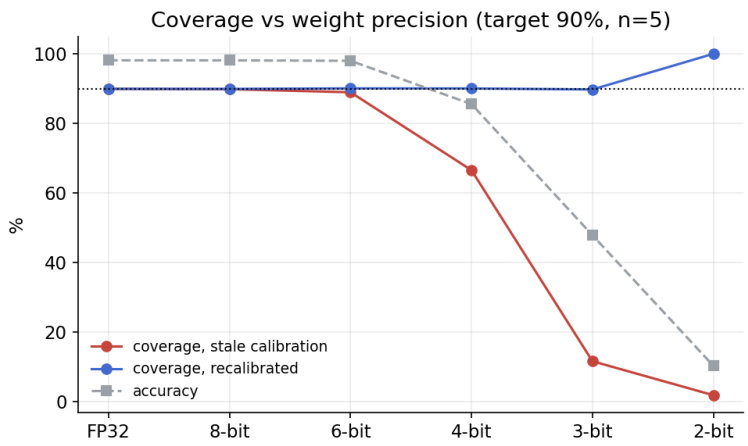


Figure 1: Coverage versus weight precision (EuroSAT/ResNet18, n=5 seeds, target 90%). Stale calibration (red) collapses smoothly; recalibration (blue) restores validity at every precision while accuracy (gray) falls.

precisely on the peaked, high-accuracy regimes typical of deployed special-purpose classifiers. A secondary observation with engineering value: MobileNetV3 is markedly more quantization-fragile than ResNet18 (deficit 0.623 vs. 0.235 at 4-bit on EuroSAT), consistent with per-channel quantization of depthwise convolutions; architecture choice is an assurance decision, not only an accuracy one.

4.4 Validity is recoverable; informativeness is not

Recalibration always restores coverage, but the guarantee can be honored vacuously. Mean prediction-set size after recalibration (Figure 4) stays near one class down to 6-bit (0.91–1.24 across the grid), passes a knee at 4-bit (1.14 for EuroSAT/ResNet18 but already 4.94 for EuroSAT/MobileNetV3), and explodes at 3-bit (5.2–9.4 classes), saturating near all 10 classes at 2-bit. Validity and informativeness thus separate cleanly under compression: the former is always repairable by re-scoring, the latter has an architecture-dependent precision floor. Mean set size on the calibration set is consequently a one-line acceptance test for whether a compressed artifact is worth deploying.

5 Deployment corollary: refresh the guarantee, not the model

Compression is one source of score shift; sensor drift over a mission is another, and the same machinery applies. Under a simulated 24-week progressive degradation (additive noise ramping to σ=0.05 and brightness to $\times$1.35; EuroSAT/ResNet18, 3 seeds, Figure 5), never-refreshed launch calibration decays from 90.7% to 20.9% coverage. Refreshing *only the conformal scores* every 4 weeks, 200 labeled samples per refresh, kilobytes of storage, no retraining or gradients, holds coverage at 81–87% through end of test; a 12-week cadence is insufficient (54.0% at week 23), so refresh rate must track drift rate. Under the working refresh, mean set size grows from 0.91 to 6.65 as model accuracy falls from 97.8% to 30.9%: the guarantee remains honest by becoming less informative, and set-size growth is precisely the onboard signal that the model, not the calibration, now requires intervention.

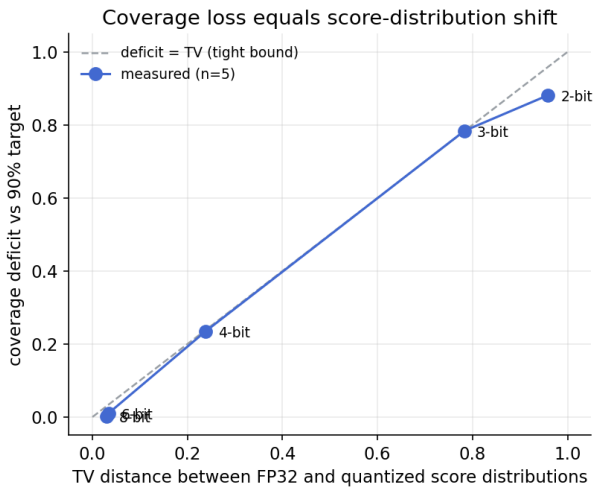


Figure 2: Coverage deficit versus TV distance between reference and deployed score distributions (EuroSAT/ResNet18). The dashed diagonal is deficit = TV.

6 Limitations

The bound in Eq. (1) is elementary and one-directional; we claim no new theory, only its instantiation, validation, and operational use for compression. TV is estimated by histograms; at extreme damage (3-bit) the estimator error is visible as a 0.001 nominal violation. Experiments cover split conformal classification with the LAC score on two mid-scale vision datasets and two architectures; regression, other scores, and larger-scale models are untested. Quantization is simulated PTQ (quantized weights, FP32 arithmetic) except where production INT8 is stated; activation quantization is not studied. Drift is synthetic and the refresh policy assumes access to small labeled batches. Findings on fault injection and input-side monitoring from the same research programme are reported in the repository rather than here.

7 Conclusion

A conformal guarantee does not survive model compression by default, but its loss is not a surprise to be discovered in deployment: it is a number that can be computed on the ground, from data already in hand, before the artifact ships. Together with the recoverability of validity, the precision knee in informativeness, and score-refresh as a maintenance policy, this yields a practical qualification recipe for compressed classifiers under distribution-free guarantees.

References

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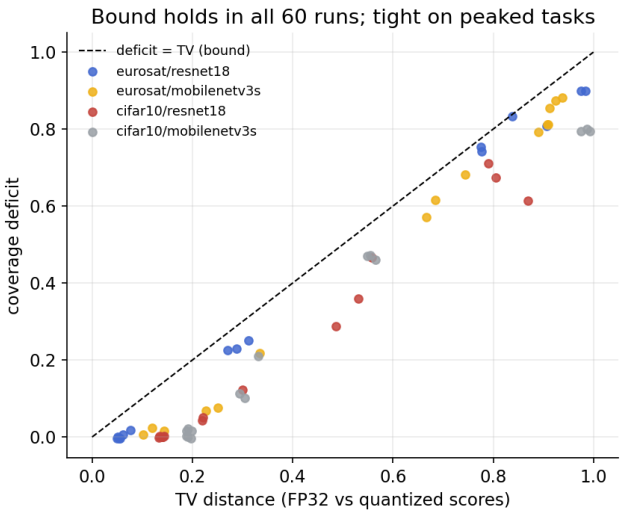


Figure 3: All 60 runs across the dataset × architecture × precision grid. Every point lies on or below the diagonal (bound holds); peaked-task points hug it.

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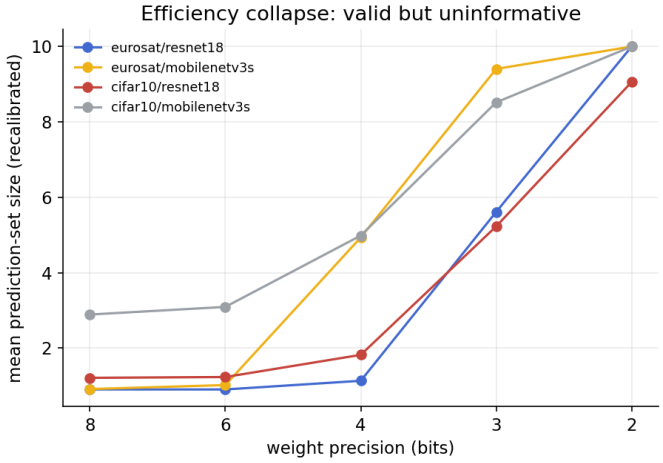


Figure 4: Mean recalibrated prediction-set size versus precision across the grid: valid but uninformative below the knee.

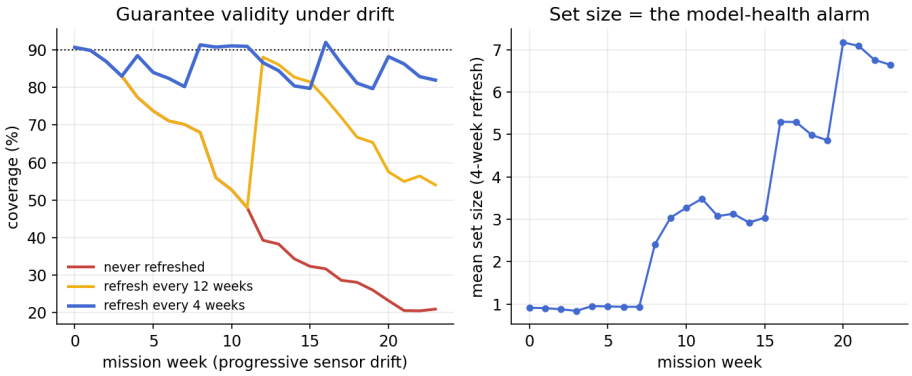


Figure 5: Simulated mission drift. Left: coverage under three maintenance policies (target 90%). Right: set size under the working policy is the model-health signal.